FEAT-ZD-48600 · FE-03 Identify the components/layers to be changed · Components, layers, sources or pipelines to change

Frontend agent · item task.components · source hobasa-frontend-agent/outputs/FEAT-ZD-48600/task.components.md · approved by automated checks and independent judge

task.components: FEAT-ZD-48600

Components and layers the Frontend lane creates or updates for FEAT-ZD-48600 (ZD-48600, "Show which overdue invoices belong to vendors that usually pay late"). This is FE-03 Identify the components/layers to be changed (attempt 5).

The Frontend scope is one task, TSK-07 (STORY-07): render the overdue risk table SC-1 with `renderOverdueRisk`. This file reads the current ap_portal code, lists what must be created or updated, and keeps the change to the smallest set of files and lines the approved design allows. Statements marked **agent judgement** are this agent's proposal; no input states them as fact. No code is changed in this step.

Each repository fact below cites the step's tool view (tool:repo), a contracted input that states it (feat.tasks, feat.stories, feat.prd, feat.trd, feat.design, task.repo_access) or record:memory.

Field	Value	Source
Trace ID	FEAT-ZD-48600	record:brief
Step	FE-03 Identify the components/layers to be changed (attempt 5)	record:brief
Task / story	TSK-07 / STORY-07, Frontend lane, repository ap_portal	feat.tasks (section 1, TSK-07); feat.stories (STORY-07)
Design component	HC-8 `renderOverdueRisk` (new); HC-9 `formatAmount`, `renderAgingSummary` (unchanged)	feat.design (A2, B4)
Repository and branch	ap_portal, branch `feature/story`, write access confirmed	task.repo_access (section 1)
Runtime / tests / CI	Node.js 22 CommonJS; `node:test` with `NODE_PATH=src`; CI command `node ci.mjs`	task.repo_access (section 2); feat.tasks (TSK-07 "Tests to add")
Code read	`src/ap_portal/index.js`, `tests/portal.test.js`	tool:repo

1. What the existing code does

Item	Current state	What it means for this change	Source		
Layer	`index.js` holds view helpers that return HTML strings ("AP portal view helpers: render the aging summary returned by the ap_aging service as HTML")	ap_portal has one product layer, view helpers. The new table is one more helper in that layer	tool:repo		
`index.js` exports	`{ VERSION, LABELS, formatAmount, renderAgingSummary }`; `VERSION = '2.3.0'`	`renderOverdueRisk` is added to this export object; the existing four exports stay	tool:repo; feat.design (B4.2)		

Item	Current state	What it means for this change	Source		
`renderAgingSummary(buckets)`	Maps LABELS keys to `<tr><td>…</td><td class="amount">…</td></tr>` with template strings, joins with `''`, wraps in `<table class="aging-summary">`	The new function follows the same plain template-string pattern and `class="amount"` cell, but must not copy the absence of escaping, because it renders feed data	tool:repo; feat.trd (F-13, IF-7.2, IF-7.3)		
`formatAmount(value)`	`Number(value \`	\	0).toFixed(2)`: formats through a binary float	Must not be reused for the overdue amount (D-2026-044); a new string-only helper is needed. `formatAmount` itself stays unchanged (TQ-2)	tool:repo; feat.trd (F-12, R-2); record:memory (D-2026-044, L-011)
Escaping	No HTML-escape helper exists	A new private helper is needed for `id` and `vendor`	tool:repo; feat.trd (F-13, IF-7.7)		
Tests	Two `node:test` tests loading the package with `require('ap_portal')`, run with `NODE_PATH=src`	New tests go in the same file with the same `require`; existing tests stay unchanged	tool:repo; feat.tasks (section 2 "Test files", TSK-07)		
Version	`2.3.0` in `src/ap_portal/index.js` and `package.json`	Both move together by one minor step, owned by TSK-07	feat.tasks (section 2 "Version step"); feat.trd (F-13)		
CI bar	Compile, unit tests, coverage of at least 90% and static checks; for ap_portal no line matching `console.log` or `eval(`, lines at most 140 characters	Every new helper branch must be exercised by a test; no `console.log`, no `eval(`, short lines	feat.tasks (section 2 "CI bar"); task.repo_access (section 2)		
Signature rule	"Keep every existing signature; add new functions."	`renderOverdueRisk` is a new function; no existing signature changes	feat.design (header, B4.1)		

2. Components and layers to create or update

ap_portal has one product layer (view helpers in `src/ap_portal/index.js`) plus its unit tests and package metadata. All new code goes into the existing `index.js`; no new source file is created (feat.design DD-2, B4.1; feat.tasks TSK-07 "Files in budget").

#	Layer	File	Component	Change	Exported	Why it is needed	Source
FC-1	View helper (constants)	`src/ap_portal/index.js`	`RISK_CLASSES = ['high', 'medium', 'low', 'unknown']`	Create	No	Allow-list for the row class, so feed data cannot inject a class name	feat.design (B4.2, B4.3); feat.trd (IF-7.4)

#	Layer	File	Component	Change	Exported	Why it is needed	Source
FC-2	View helper (constants)	`src/ap_portal/index.js`	`OVERDUE_HEADINGS = ['Invoice', 'Vendor', 'Days overdue', 'Amount']`	Create	No	The four headings in order; "Days overdue", never "Late"	feat.design (B4.2, NF-12); feat.trd (IF-7.2); feat.prd (AC-19, NFR-10)
FC-3	View helper (private)	`src/ap_portal/index.js`	`escapeHtml(value)`	Create	No	Escapes &, <, >, ", ' in `id` and `vendor`	feat.design (B4.2, SE-2); feat.trd (IF-7.7)
FC-4	View helper (private)	`src/ap_portal/index.js`	`toTwoDecimalString(value)`	Create	No	String-only amount display: accepts only a string matching `^-?\d+(\.\d{1,2})?$`, pads the fraction to two digits as text, else throws `TypeError`; never `Number()`, `parseFloat` or `formatAmount`	feat.design (B4.2, B4.4, DD-5, DD-8); feat.tasks (TSK-07 "Implementation notes")
FC-5	View helper (private)	`src/ap_portal/index.js`	`riskClass(risk)`	Create	No	`'risk-' + risk` for a value in `RISK_CLASSES`, else `'risk-unknown'`	feat.design (B4.2); feat.trd (IF-7.4)
FC-6	View helper (public)	`src/ap_portal/index.js`	`renderOverdueRisk(rows)`	Create	Yes	Renders SC-1 from CT-1 rows: `''` for missing, `null` or `[]`; `TypeError` for any other non-array; `<table class="overdue-risk">`, plain header row, one `<tr class="risk-…">` per row in the given order with cells id, vendor, days, amount (`class="amount"`); `days` must pass `Number.isInteger`, else `TypeError`	feat.design (HC-8, B4.2 steps 1 to 4, B4.4); feat.tasks (TSK-07 "Goal")

#	Layer	File	Component	Change	Exported	Why it is needed	Source
FC-7	Module interface	`src/ap_portal/index.js`	`module.exports`	Update (additive)	—	Add `renderOverdueRisk` next to `VERSION`, `LABELS`, `formatAmount`, `renderAgingSummary`	feat.design (B4.2); feat.stories (AC-07.10); tool:repo
FC-8	Package metadata	`src/ap_portal/index.js`	`VERSION`	Update	(already exported)	One minor version step from `2.3.0`, moved together with `package.json`; exact number for the lane to confirm at coding	feat.tasks (section 2 "Version step", TSK-07 "Done when"); feat.design (DD-11)
FC-9	Package metadata	`package.json`	`"version"`	Update (version only)	—	Same value as `VERSION`	feat.tasks (TSK-07 "Files in budget", TSK-07 "Done when")
FC-10	Unit tests	`tests/portal.test.js`	New `node:test` cases	Update (additive)	—	Cover FC-1 to FC-7 and every error branch for the 90% coverage bar; existing two tests unchanged	feat.design (B4.5); feat.tasks (section 2 "CI bar", TSK-07 "Tests to add"); tool:repo

```
flowchart LR
  CT1["CT-1 rows (JSON.parse of ap_aging payload)"] --> R["renderOverdueRisk (FC-6, exported)"]
  R --> RC["riskClass (FC-5)"] --> AL["RISK_CLASSES (FC-1)"]
  R --> EH["escapeHtml (FC-3)"]
  R --> TD["toTwoDecimalString (FC-4)"]
  R --> HD["OVERDUE_HEADINGS (FC-2)"]
  R --> HTML["&lt;table class=overdue-risk&gt; or ''"]
  FA["formatAmount (unchanged, not used)"]
  RA["renderAgingSummary (unchanged)"]
```

Sources: feat.design (A3 CT-1, B4.1, B4.2); tool:repo.

3. Components that stay unchanged

Component	File	Rule	Source
`LABELS`	`src/ap_portal/index.js`	No change	feat.tasks (TSK-07 "Must not change"); feat.design (B4.1)
`formatAmount`	`src/ap_portal/index.js`	No change and not called by the new code (float conflict kept open as TQ-2)	feat.trd (F-12, C-9, TQ-2); feat.design (HC-9, DD-5)
`renderAgingSummary`	`src/ap_portal/index.js`	No change and not reused	feat.trd (C-9); feat.design (HC-9)
Existing tests (`formats amounts with two decimals`, `renders every bucket in order`)	`tests/portal.test.js`	No change; must still pass	tool:repo; feat.stories (AC-07.10); feat.trd (T-16)

Component	File	Rule	Source
Any file outside the budget	—	No change; the budget is `index.js`, `package.json` (version only) and `tests/portal.test.js`	feat.tasks (TSK-07 "Files in budget"; section 2 "Size budget")
Transport, host page, styles, row actions	—	Not in ap_portal and out of this design (no transport layer; no host page, navigation, filters or row actions)	feat.design (A1, DD-3, A8); feat.stories (AC-07.9)

4. Minimum-code approach

The approved design already fixes the component list (FC-1 to FC-7). Within it, this step proposes how to keep the change small (agent judgement unless a source is given):

1. **One file, appended.** Add the two constants and four functions to `index.js` between `renderAgingSummary` and `module.exports`; change only the `VERSION` line and the export line among existing lines. No new module, so the `require('ap_portal')` resolution and the test setup need no change (feat.design B4.1; tool:repo).
2. **Reuse the existing pattern, not the existing helpers.** Build rows with `map(...).join('')` and template strings as `renderAgingSummary` does; do not call `formatAmount` (tool:repo; feat.design DD-5).
3. **Header from the constant.** Build the header row by mapping `OVERDUE_HEADINGS` to `<th>` cells and joining them with `''` inside a plain `<tr>`, so the four headings appear once in the code (feat.trd IF-7.2).
4. **Single validation point per field.** `toTwoDecimalString` owns all amount checks (type and pattern in one regular-expression test); the `days` check is one `Number.isInteger` test inside the row template; `riskClass` is one `includes` lookup. Each throw path is short, which keeps the branches few and easy to cover (feat.design B4.4; feat.tasks TSK-07 "Implementation notes").
5. **Padding as text.** Split the matched string at `.` and pad the fraction with `'0'` as text; no number conversion (feat.design B4.2; feat.stories AC-07.5, AC-07.6).
6. **No new dependency, no build step.** Node.js 22 CommonJS built-ins only (feat.design DD-1).
7. **Static-check safe names.** No identifier or comment contains `console.log` or `eval(`; lines stay at most 140 characters (feat.tasks section 2 "CI bar").

5. Tests to add (FC-10)

All in `tests/portal.test.js`, using `require('ap_portal')` and synthetic data (`SYN-INV-*`, "Synthetic Vendor <letter>"). Private helpers are covered through `renderOverdueRisk`.

Test	Covers	Acceptance criteria / TRD test	Source
`renders overdue risk table`	FC-2, FC-6, FC-7	AC-07.1, AC-07.2; T-12	feat.design (B4.5)
`maps every risk value to its class`	FC-1, FC-5	AC-07.3; T-12	feat.design (B4.5)
`returns empty string for no rows` (`[]`, `undefined`, `null`)	FC-6	AC-07.4; T-13	feat.design (B4.5); feat.tasks (TSK-07 "Tests to add")
`shows amount strings unchanged` (`"1250.00"`, `"15.00"`, a string with more digits than a JS `Number` holds)	FC-4	AC-07.5; T-14	feat.design (B4.5)
`pads short amounts as strings` (`"980.5"` → `980.50`)	FC-4	AC-07.6; T-14	feat.design (B4.5)

Test	Covers	Acceptance criteria / TRD test	Source
`rejects non-string amounts`; exponent, thousands separator, three decimals → `TypeError`	FC-4 error branch	DD-8	feat.design (B4.5); feat.tasks (TSK-07 "Tests to add")
`escapes id and vendor` (<, &, ")	FC-3	AC-07.7; T-15	feat.design (B4.5)
`unexpected risk renders risk-unknown`	FC-5	AC-07.8; T-15	feat.design (B4.5)
Non-array `rows` → `TypeError`; non-integer `days` → `TypeError`	FC-6 error branches	DD-8	feat.tasks (TSK-07 "Tests to add")
Existing tests unchanged	`LABELS`, `formatAmount`, `renderAgingSummary`	AC-07.10; T-16	tool:repo; feat.trd (T-16)

6. Traceability

Acceptance criterion (STORY-07)	Components
AC-07.1 table, headings, one row per item in order	FC-2, FC-6
AC-07.2 plain header row [Pending TQ-7]	FC-2, FC-6
AC-07.3 risk classes	FC-1, FC-5
AC-07.4 empty string for no rows	FC-6
AC-07.5 amount strings unchanged, no float	FC-4
AC-07.6 padding [Pending TQ-11, OQ-B10]	FC-4
AC-07.7 HTML escaping [Pending OQ-B8]	FC-3
AC-07.8 unknown risk → `risk-unknown` [Pending TQ-6]	FC-1, FC-5
AC-07.9 no Risk column, no row actions [Pending OQ-B9, PQ-P7]	FC-2, FC-6 (nothing added)
AC-07.10 existing exports unchanged, new export added, no `console.log` / `eval`	FC-7, FC-10; section 3

Sources: feat.stories (STORY-07); feat.design (B4.5, D1); feat.trd (IF-7, T-12 to T-16).

7. Assumptions (agent judgement, for the gate)

ID	Assumption
CA-1	All new code lives in `index.js`; no new module is created, so the CI setup and package resolution stay as they are
CA-2	Private helpers are not exported and are tested only through `renderOverdueRisk`
CA-3	The `risk-*` classes need no stylesheet change in ap_portal. The tool view shows only view helpers that return HTML strings, and the host page is out of this design
CA-4	The interim answers in feat.design (TQ-6 allow-list, TQ-7 plain header, TQ-11 padding, OQ-B8 escaping, DQ-3 `TypeError`) are built as designed; if an answer differs, only FC-1/FC-5, FC-2, FC-4, FC-3 or the throw paths change

8. Open questions (all non-blocking)

ID	Question	Interim answer	Owner (proposed)
CQ-FE-1	Where are the `risk-high`, `risk-medium`, `risk-low` and `risk-unknown` styles defined? ap_portal only emits the class	Outside ap_portal; the class is the only output (CA-3)	UX lead
CQ-FE-2	Confirm the exact next minor version for `VERSION` and `package.json`	Next minor step from `2.3.0`, set during coding (agent judgement)	Developer
RAQ-1 (carried)	Is `feature/story` the intended branch for TSK-07?	Work on `feature/story`; commits start with `STORY-07`	Developer

Carried, unchanged: TQ-2, TQ-6, TQ-7, TQ-11 (feat.trd); OQ-B8, OQ-B9, OQ-B10 (feat.prd); DQ-3 (feat.design); PQ-P7 (feat.prototype, as carried in feat.stories). None blocks this step.

9. Validation check

Check	Result	Source
Existing code read before listing changes	Pass (section 1)	tool:repo
Every component traces to the design and a story criterion	Pass (sections 2, 6)	feat.design (B4); feat.stories (STORY-07)
Only files in the TSK-07 budget are changed	Pass: `index.js`, `package.json` (version only), `tests/portal.test.js`	feat.tasks (TSK-07)
Existing signatures kept; new function added	Pass	feat.design (header, B4.1)
D-2026-044 / L-011 (no float for money)	Consistent: FC-4 is string-only; `formatAmount` not used	record:memory; feat.design (NF-1)
Blocking questions	None	—